

SYSTEM ID REPORT – PHASE 1

(HEXAPOD ROBOT)

1. SYSTEM DESCRIPTION

The system is a hexapod walking robot consisting of six legs, each driven by multiple servo motors. The robot is controlled using an Arduino Uno interfaced with PCA9685 servo driver modules via the I2C protocol. The system enables stable locomotion using coordinated gait patterns such as tripod gait and joystick-based movement.

The primary objective is to achieve controlled walking, turning, and directional movement using manual input (joysticks).

2. INPUT VARIABLES (I)

- Joystick 1:
 - X-axis → Left / Right movement
 - Y-axis → Forward / Backward movement
- Joystick 2:
 - X-axis → Rotation (Turning)
 - Y-axis → Height / Posture adjustment (optional)
- Power Supply Input (Battery voltage)

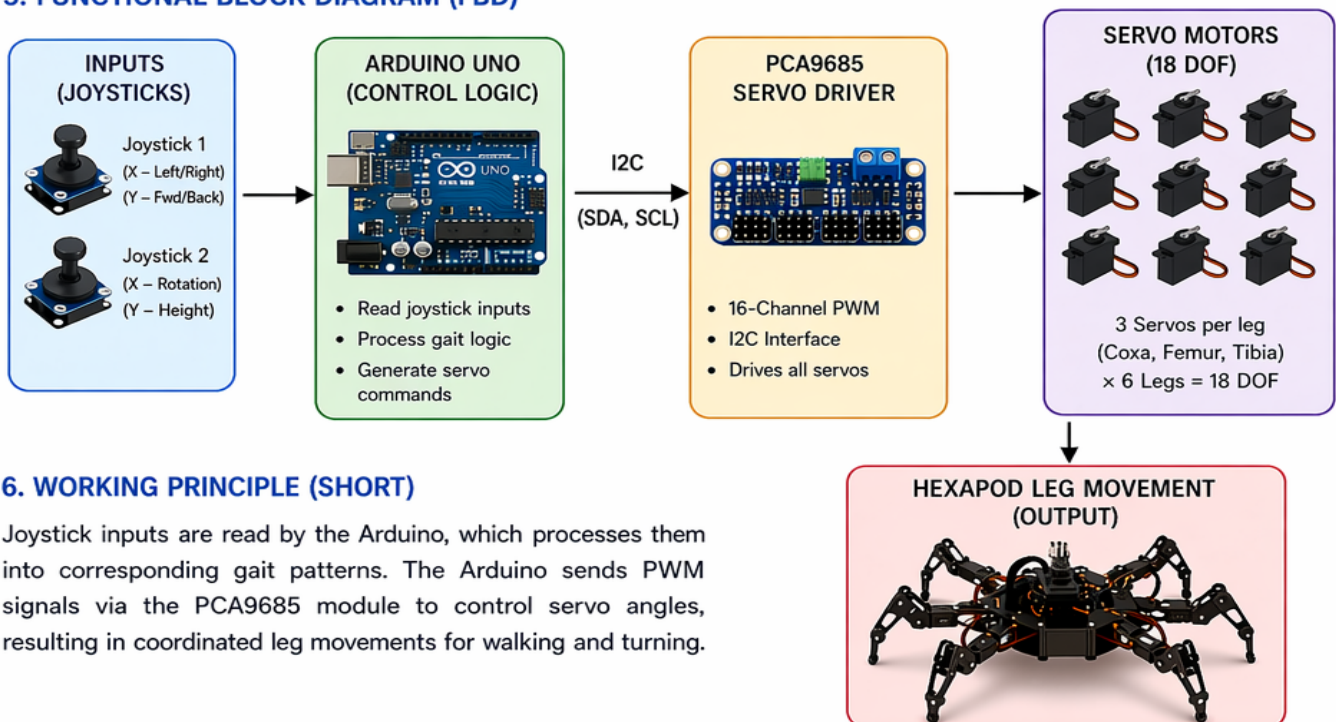
3. OUTPUT VARIABLES (O)

- Servo motor angles (0° – 180°) for each joint:
 - Coxa (horizontal movement)
 - Femur (lifting)
 - Tibia (extension)
- Robot motion:
 - Walking (forward / backward)
 - Turning (left / right)
 - Stability adjustments

4. ASSUMPTIONS

- Each servo operates within a 0° to 180° range
- Power supply is sufficient (typically 5–6V stable supply for servos)
- Communication via I2C (SDA, SCL shared) is reliable
- No significant delay in servo response
- Terrain is assumed to be flat during initial testing
- Weight distribution is balanced across all six legs

5. FUNCTIONAL BLOCK DIAGRAM (FBD)



6. WORKING PRINCIPLE (SHORT)

Joystick inputs are read by the Arduino, which processes them into corresponding gait patterns. The Arduino sends PWM signals via the PCA9685 module to control servo angles, resulting in coordinated leg movements for walking and turning.